

Conservation of Energy

Name: _____

Date: _____

A. Lesson Recap

- 1) Energy is the ability to do _____.
- 2) Mechanical energy is energy associated with _____.
- 3) List the three forms of mechanical energy:

_____, _____, _____

- 4) Why do we say “at that instant” when identifying energy?

B. Energy Systems and Conservation

- 1) Energy within an open system is not conserved.
 - 2) Energy within a closed system is conserved.
- 5) If energy is conserved in a closed system, what can still happen to energy? (Circle all that apply)
- a) Energy can change form.
 - b) Energy can disappear.
 - c) Energy can transfer from one type to another.
 - d) The total amount must stay the same.

To conserve energy, whatever energy we started with in the beginning, must end with the same amount of energy.

This can be modelled using the following formula:

$$\sum E_i = \sum E_f$$

C. Mechanical Energy vs Total Energy

Mechanical Energy includes:

$$K \text{ (KE)}, \quad U_g \text{ (PE)}_g, \quad U_s \text{ (PE)}_s$$

Mechanical energy assumes no loss of energy, which means that any system we deal with, there will be no air resistance and surfaces will be frictionless.

Total Energy includes:

$$K \text{ (KE)}, \quad U_g \text{ (PE)}_g, \quad U_s \text{ (PE)}_s \quad \& \quad Q \text{ (TH)}$$

Total energy can be modelled using the following formula:

$$E_{\text{total}} = E_{\text{mech}} + E_{\text{th}}$$

When some energy becomes thermal energy, it cannot be reused in the mechanical-energy model.

6) What common physical situation causes thermal energy to increase?

D. The Full Formulas

Mechanical Energy Formula

$$E_{\text{mech}} = K + U_g + U_s$$

Total Energy Formula

$$E_{\text{total}} = (K + U_g + U_s) + E_{\text{th}}$$

Conservation of Energy Formula (Expanded)

$$\begin{aligned} \sum E_i &= \sum E_f \\ \sum E_{i_{\text{mech}}} &= \sum E_{f_{\text{mech}}} \end{aligned}$$

For mechanical energy (no thermal transfer), we can write:

$$K_i + U_{g,i} + U_{s,i} = K_f + U_{g,f} + U_{s,f}$$

Conservation of Total Energy (With Thermal)

If energy transfers into thermal energy, we model:

$$K_i + U_{g,i} + U_{s,i} + E_{\text{th},i} = K_f + U_{g,f} + U_{s,f} + E_{\text{th},f}$$

E. Pie Chart Connection

Energy pie charts model parts of a total.

7) If mechanical energy is conserved, how does the *total size of the pie* change (or not change)?

8) If mechanical energy is *not* conserved because friction is present, what happens to the pie chart idea? (Where does the “missing” mechanical energy go?)

F. Practice

Directions: For each scenario:

1. Define the **system**.
 2. Choose a **reference point**.
 3. Identify energy types at three instants.
 4. Decide: Is **total energy** conserved? Is **mechanical energy** conserved?
 5. Complete additional prompts only when provided.
1. A ball is dropped from rest from height h . (Ignore air resistance)
- 1) What is the system?
 - 2) What is the reference point in which ($h = 0$ m)?
 - 3) What is the energy distribution present in each of the following instants:

- Release: K U_g U_s E_{th}

- Midway: $K \quad U_g \quad U_s \quad E_{th}$
- Just before impact: $K \quad U_g \quad U_s \quad E_{th}$

4) Is energy conserved? Total Energy: Yes / No Mechanical Energy: Yes / No

5) Draw the energy pie charts for each instant:

6) Write the expanded conservation equation for this scenario:

7) Explain why mechanical energy is conserved in this case.

2. A block slides on a rough horizontal surface and slows down. (Include friction)

1) What is the system?

2) What is the reference point in which ($h = 0$ m)?

3) What is the energy distribution present in each of the following instants:

- Moving fast: $K \quad U_g \quad U_s \quad E_{th}$
- Moving slower: $K \quad U_g \quad U_s \quad E_{th}$
- Stopped: $K \quad U_g \quad U_s \quad E_{th}$

4) Is energy conserved? Total Energy: Yes / No Mechanical Energy: Yes / No

5) Draw the energy pie charts for each instant:

6) Write the expanded conservation equation for this scenario:

7) Identify which energy term increases and explain why.

3. A skateboarder rolls down a smooth hill and back up the other side. (Ignore friction)

1) What is the system?

2) What is the reference point in which ($h = 0$ m)?

3) What is the energy distribution present in each of the following instants:

• Top of first hill:	K	U_g	U_s	E_{th}
• Bottom of hill:	K	U_g	U_s	E_{th}
• Top of opposite hill:	K	U_g	U_s	E_{th}

4) Is energy conserved? Total Energy: Yes / No Mechanical Energy: Yes / No

5) Write the expanded conservation equation for this scenario:

6) Predict whether the skateboarder reaches the same height on the opposite side and justify your answer.

4. A pendulum swings back and forth without air resistance.

1) What is the system?

2) What is the reference point in which ($h = 0$ m)?

3) What is the energy distribution present in each of the following instants:

• Highest point:	K	U_g	U_s	E_{th}
• Lowest point:	K	U_g	U_s	E_{th}
• Opposite highest point:	K	U_g	U_s	E_{th}

4) Is energy conserved? Total Energy: Yes / No Mechanical Energy: Yes / No

5) Explain why the pendulum never exceeds its starting height.

5. A car brakes to a stop on a level road.

1) What is the system?

2) What is the reference point in which ($h = 0$ m)?

3) What is the energy distribution present in each of the following instants:

• Moving fast:	K	U_g	U_s	E_{th}
• Slowing down:	K	U_g	U_s	E_{th}
• Stopped:	K	U_g	U_s	E_{th}

4) Is energy conserved? Total Energy: Yes / No Mechanical Energy: Yes / No

5) Explain how the brakes increase thermal energy.

6. A block slides and compresses a spring until it comes to rest. (No friction)

1) What is the system?

2) What is the reference point in which ($h = 0$ m)?

3) What is the energy distribution present in each of the following instants:

• Before contact with spring:	K	U_g	U_s	E_{th}
• During compression:	K	U_g	U_s	E_{th}
• Maximum compression:	K	U_g	U_s	E_{th}

4) Is energy conserved? Total Energy: Yes / No Mechanical Energy: Yes / No

5) Draw the energy pie charts for each instant:

6) Write the expanded conservation equation for this scenario:

7) Identify which potential energy term stores the initial kinetic energy.